MODULE 5 SPINAL CORD

The Spinal Cord.

THE CENTRAL NERVOUS SYSTEM, THE SPINAL CORD

Covers the external anatomy of the cord, its enlargements and terminations, the roots and rootlets that connect it to the spinal nerves, and the internal anatomy of its gray and white matter. The focus is on the structures and the job each one does.

BY THE END

1. Describe the external anatomy of the spinal cord, including its enlargements and terminations.
2. Identify the roots, rootlets, and dorsal root ganglion.
3. Describe the internal anatomy of the gray horns and the white columns and tracts.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The spinal cord

drsrennie-stack.github.io/new-build-bio4-solano/spinal-cord-concept-videos.html

The Spinal Cord, an Overview

The spinal cord runs from the medulla oblongata down the vertebral canal. It carries signals between the brain and the body and is the center for spinal reflexes.

THE CORD, ITS DIMENSIONS, AND WHAT IT DOES

Structure	What it is
Spinal cord	The cord of nervous tissue within the vertebral canal, continuous with the medulla above
Length	About 16 to 18 inches in the adult, running from the medulla to the L1 to L2 level
Diameter	Not uniform; widest in the cervical region and narrowest in the thoracic region
Cord growth	The cord stops lengthening at about 4 to 5 years of age, while the vertebral column keeps growing
Functions	Conducting sensory and motor signals between the brain and the body, and acting as the integration center for spinal reflexes

External Anatomy of the Spinal Cord

Two regions of the cord are visibly wider, and the cord ends well above the end of the vertebral column. Two grooves run its full length.

THE ENLARGEMENTS, THE TERMINATIONS, AND THE GROOVES

Structure	What it is
Cervical enlargement	A widening from C4 to T1 where the nerves of the upper limbs arise
Lumbar enlargement	A widening behind about the T9 to T12 vertebrae where the nerves of the lower limbs arise
Conus medullaris	The tapered, cone-shaped end of the spinal cord, at the disc between L1 and L2
Filum terminale	A thread of pia mater extending from the conus medullaris that anchors the cord to the coccyx
Cauda equina	The hair-like bundle of nerve roots that descends past the conus medullaris within the vertebral canal
Anterior median fissure	The deep groove down the front of the cord
Posterior median sulcus	The shallow groove down the back of the cord; with the fissure it divides the cord into right and left halves

The Roots and Rootlets

Each side of the cord connects to a spinal nerve through two roots. Tiny rootlets gather into a root, and the two roots join to form the mixed spinal nerve.

FROM ROOTLET TO SPINAL NERVE

Structure	What it is
Rootlets	The small bundles of axons that attach in a row along the cord and merge to form a root
Root	A bundle of axons that connects a spinal nerve to the cord; each spinal nerve has two
Posterior (dorsal) root	The sensory root; it carries sensory axons from receptors in the periphery into the cord
Posterior (dorsal) root ganglion	The swelling on the posterior root that holds the cell bodies of the sensory neurons
Anterior (ventral) root	The motor root; it carries motor axons from the cord out to muscles and glands
Spinal nerve	The mixed nerve formed where the posterior and anterior roots join; it carries both sensory and motor axons
Intervertebral foramen	The gap between adjacent vertebrae where the spinal nerve exits the vertebral canal

KEY POINT

Posterior is sensory in, anterior is motor out, and only the posterior root carries a ganglion. If a question puts cell bodies on a root, it is the posterior one.

Internal Anatomy: The Gray Matter

A cross-section of the cord shows a butterfly of gray matter at the core, wrapped in white matter. This is the reverse of the brain, where gray matter is on the outside.

THE GRAY CORE AND ITS BRIDGES

Structure	What it is
Gray matter	The inner, butterfly-shaped region of neuron cell bodies
Central canal	The small cerebrospinal fluid channel at the center of the cord, continuous with the fourth ventricle
Gray commissure	The bridge of gray matter that connects the two sides, surrounding the central canal
Anterior white commissure	The bridge of white matter that connects the two sides of the cord

The gray matter is shaped into projections called horns. Compare what each horn holds.

THE GRAY HORNS AND THEIR CONTENTS

Horn	Contents
Posterior (dorsal) horn	Interneurons and the incoming axons of sensory neurons
Lateral horn	Autonomic motor nuclei; present only in the thoracic and upper lumbar cord, T1 to L2
Anterior (ventral) horn	Somatic motor nuclei that drive skeletal muscle contraction

Internal Anatomy: The White Matter

The white matter of the cord is sorted into columns, and each column holds tracts. A tract is a bundle of axons that share a common origin and destination.

THE COLUMNS AND THE TRACTS THEY HOLD

Structure	What it is
White matter	The outer region of the cord, made of myelinated axons
Columns	The anterior, posterior, and lateral white columns, the major divisions of the cord's white matter
Tracts	Bundles of axons within the columns that share a common origin and destination
Ascending tracts	Sensory tracts that conduct impulses up toward the brain; also called afferent tracts
Descending tracts	Motor tracts that conduct impulses down away from the brain; also called efferent tracts

A cluster of cell bodies and a bundle of axons each have a different name in the CNS and the PNS. Compare the four terms.

THE FOUR TERMS BY LOCATION

Location	Cluster of cell bodies	Bundle of axons
Central nervous system	Nucleus	Tract
Peripheral nervous system	Ganglion	Nerve

Distinguishing the Cord Segments

A cross-section taken at different levels of the cord does not look the same. The proportions of gray and white matter, the shape, and the size all shift from top to bottom. Compare them.

TELLING THE CORD SEGMENTS APART IN CROSS-SECTION

Segment	Distinguishing features
Cervical	Large diameter with a large amount of white matter; oval in shape
Thoracic	Small diameter with a small amount of gray matter; has a small lateral gray horn
Lumbar	Almost circular; less white matter than the cervical segments
Sacral	Relatively small, with more gray matter and little white matter
Coccygeal	The smallest; resembles the lower sacral segments

KEY POINT

White matter thins as you go down the cord, because fewer tracts are still running, while gray matter grows where a limb needs serving. That is why a cervical section is mostly white and a sacral section is mostly gray.

WHERE THE CORD ENDS, AND WHAT THAT ALLOWS

The spinal cord ends at the **conus medullaris** near the L1 to L2 disc, but the vertebral column runs much lower. Below the conus, only the loose **cauda equina** hangs in the canal. That gap is the safe target for a lumbar puncture: a needle can draw cerebrospinal fluid there without risking the cord itself.

The **posterior root ganglion** is the home address of every sensory neuron cell body for that segment. The varicella virus hides in these ganglia for decades; when it reactivates as shingles, the rash follows the exact strip of skin served by that one ganglion.

A spinal cord injury interrupts the **ascending** and **descending tracts**. Because the tracts carry every signal between the brain and the body below, function is lost below the level of the injury, while everything above it is spared.